

Silent Vision

Closed-set visual command recognition from a short, silent camera clip.

PROJECT PROFILE

Silent control when audio is not an option.

The browser records a 2-5 second WebM clip without audio. The backend extracts a mouth-region sequence, checks confidence and margin, and returns a structured command decision plus an agent result.

Bounded vocabulary

LIGHT_ON, LIGHT_OFF, OPEN_DOOR, CHAT_OTHER, and UNKNOWN. Silent Vision does not claim open-ended transcription.

Inspectable result

Candidates, confidence, margin, acceptance reason, and execute, ignore, or reject are returned as structured data.

CURRENT STATUS

The proof of concept does not control a physical light or door and does not create a browser recording or still-image artifact. The intended demo uses CPU preprocessing and a PyTorch temporal classifier on AMD Radeon through ROCm.

PENDING

Final Radeon checkpoint, validation evidence, and recorded demonstration.

Target users

A deliberate visual input for situations where audio is unavailable, unreliable, or unwanted.

Accessible service

A Deaf or hard-of-hearing person may need a visual alternative at a service desk.

Creator studio - planned

A creator may need hands-free control near a loud set. Browser recording and still capture are not implemented.

Noisy worksite

An operator may work around machinery where speech recognition is unreliable.

Privacy-sensitive team

A deliberate camera gesture can replace an always-listening microphone for a small command vocabulary.

PRODUCT BOUNDARY

Silent Vision answers "which allowed command best fits this clip?" It does not claim a transcript. An integrator decides exactly which accepted labels may reach a downstream system.

The current smart-space labels demonstrate that interface boundary. Creator-control actions remain planned work.

Product workflow and safety

One short recording, one bounded decision, and explicit uncertainty handling.

01

Start

User grants camera access; audio remains off.



02

Prepare

A countdown gives the user time to frame the command.



03

Record

The browser records one WebM clip for up to five seconds.



04

Classify

The server returns candidates, confidence, margin, and a reason.



05

Route

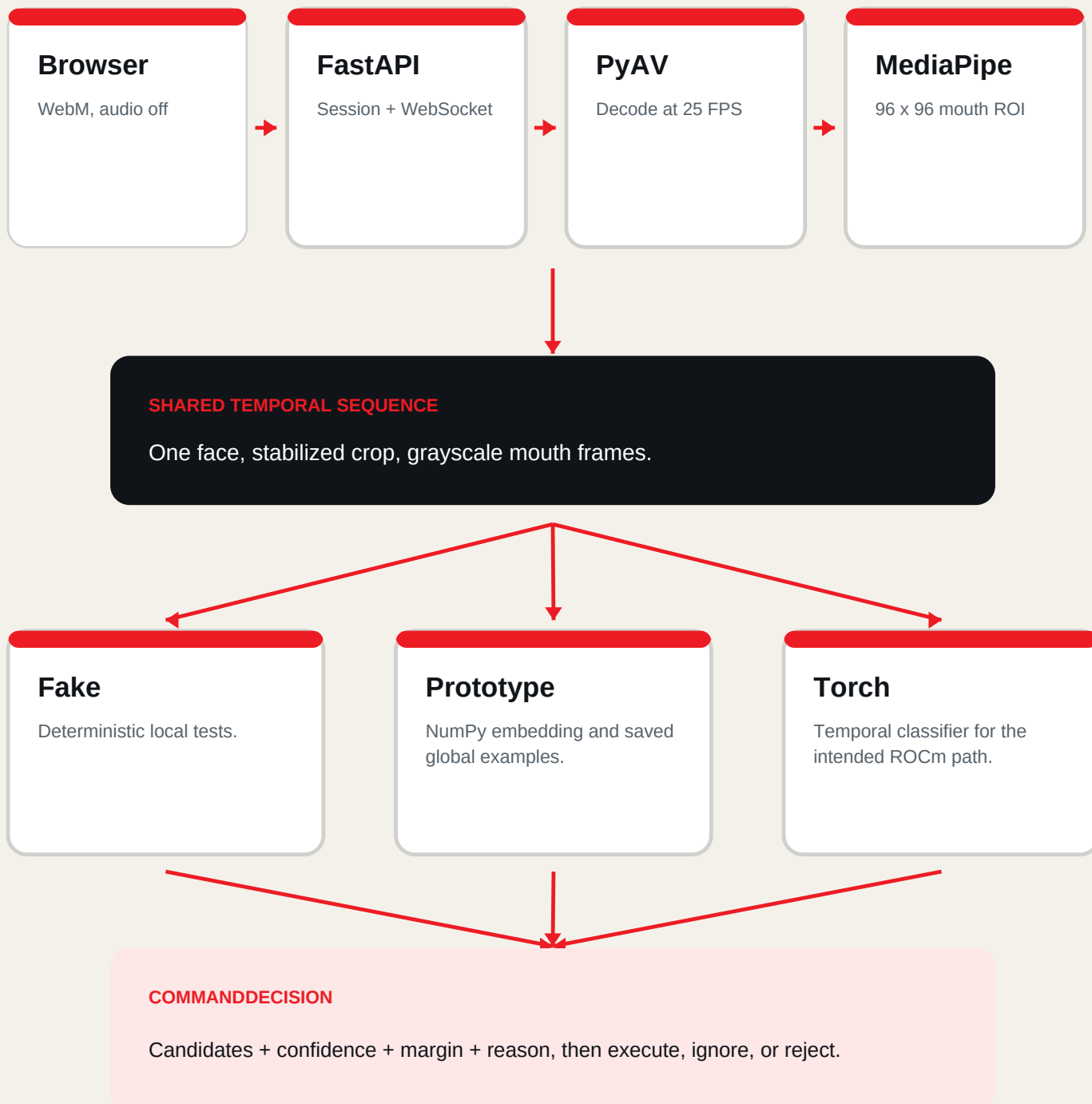
The agent boundary returns execute, ignore, or reject.

SAFETY RULE

Low confidence, a small top-1/top-2 margin, or an UNKNOWN prediction produces a rejection. CHAT_OTHER can be recognized but non-executable.

System architecture

A CPU media pipeline feeds interchangeable command backends with one decision schema.



Model and algorithm

Prototype calibration and Torch classification share the same rejection boundary.

1

Frame features

Mean, standard deviation, and motion values



2

4 temporal blocks

Feed-forward, attention, depthwise 1D convolution, normalization



3

Attentive pooling

Reduce the temporal sequence



4

Label classifier

Score the bounded vocabulary

Prototype mode

Normalizes an appearance-and-motion embedding, compares cosine similarity, then checks confidence and margin. It is a calibration tool, not Radeon evidence.

Training status

Scripts exist for train, validate, and infer. Real mouth_roi_npy rows and a separate held-out validation manifest are still required.

AMD Radeon and ROCm

Intended demo path: CPU preprocessing, then PyTorch temporal classification on Radeon.

CPU preprocessing

PyAV decoding, MediaPipe detection, crop stabilization, and NumPy feature work stay on CPU.

Radeon classification

PyTorch runs the temporal classifier through ROCm. The ROCm-backed device appears as cuda:0 and torch.version.hip records HIP availability.

REPRODUCTION GUARDS

- HIP must be present.
- The accelerator must be available.
- A Torch checkpoint path must be provided.

EVIDENCE STATUS

Pending: final Radeon checkpoint, held-out validation report, selected environment record, Creator Mode actions, and the 3-5 minute end-to-end video. No accuracy or latency number is published.

NEXT STEPS

1. Finish a balanced dataset. 2. Build a real manifest. 3. Train on Radeon. 4. Evaluate a held-out manifest. 5. Save reports and record the demo.

<https://github.com/fangjixin/silent-vision>